

BMGT 302 Essential Programming and AI Skills for Business Analytics

Fall 2026 - Syllabus

Lecture Meeting Time	Section 0401 – Monday & Wednesday – 8:00-9:15am
Lecture Location	Van Munching Hall (VMH) 2203
Course Website	ELMS/Canvas
Prerequisite	BMGT301, or permission of the Robert H. Smith School of Business
Instructor	Xiang (Charlie) Cheng
Questions & Inquiries	ELMS/Canvas Inbox for anything personal, ELMS discussion board for course content (do NOT email)
Office Hour	By appointment
Office Hour Location	By appointment on Zoom

COURSE OVERVIEW

This course introduces structured and object-oriented programming, focusing on their applications in business functions and analytics. Students apply problem-solving skills to create software solutions using programming logic and data structures. The course emphasizes learning by doing through practical exercises with modern programming languages, and covers AI coding assistants with prompt engineering techniques.

MATERIALS

No textbook purchase is required. Both books below are optional.

Recommended Python Textbook:	Recommended Practical Textbook:
Python for Everybody (3rd Edition)	Automate the Boring Stuff with Python (2nd Edition)

Recommended Python Textbook:	Recommended Practical Textbook:
Author: Charles R. Severance	Author: Al Sweigart
ISBN: 9781530051120 for the optional print edition	ISBN: 9781593279929 for the optional print edition

TECHNOLOGY REQUIREMENT

- A laptop, brought to every session
- Google Colab – free with your UMD Google account, used in Units 1 and 2
- Python 3.12 and Visual Studio Code – free, installed in Session 1 and used from Unit 3
- Microsoft Excel, desktop version – free through UMD, used in Sessions 17 and 18
- Git – free, used from Session 22 and run locally
- Microsoft Copilot and Google Gemini – free through UMD
- An API key for the language-model unit – issued by the course

If a technology requirement creates a hardship, including a machine you are not able to install software on, contact me in the first two weeks.

LEARNING OBJECTIVES

This is a demanding course. The objective is to equip you with the knowledge and skills to use programming as a tool in solving business problems, and to judge whether code an AI assistant produces does what the business asked for.

Upon completion of this course, the student should be able to:

1. Decompose a business problem into a sequence of operations a computer carries out.
2. Read a Python program and predict its output before executing it.
3. Write Python programs using variables, conditionals, loops, functions, and classes.
4. Load, clean, combine, and summarize business data with pandas.
5. Automate spreadsheet work from Python, and work out what inherited VBA does.
6. Specify a programming task precisely enough that an AI assistant returns usable code.
7. Verify AI-generated code, locate where it is wrong, and correct it.
8. Call a language model from a program to process business text at scale.
9. Estimate the cost, the return, and the failure modes of an AI system before building it.

10. Improve working code without changing its behavior, using an AI assistant and version control.

Objectives 1 through 5 cover the fundamentals. Objectives 6 through 10 cover working with an assistant.

LECTURES

The lectures will focus on discussing concepts with simple examples. The lecture meetings will be in the classroom. The student will be responsible for all the material and updates presented and discussed in the lectures. The lecture notes and lectures will provide a base of knowledge that participants can use in assignments and the course project.

GRADING

The Smith School of Business offers rigorous, academically challenging courses and provides meaningful feedback on student performance to facilitate learning. Transparency and consistency in grading are important elements that ensure the integrity of the curriculum.

Grading Method

Component	Weight	How it is counted
In-Class Activities (attendance)	20%	One activity in each of the 27 sessions, scored 0 or 1 for a submitted attempt. The 5 lowest scores are dropped.
Individual Assignments	20%	5 homework assignments, equal weight
Course Project	30%	Two graded milestones plus an ungraded project brief, built individually across the semester
Final Exam	30%	On paper, open-note, closed-device
Total	100%	

Final Grades

This course uses the plus/minus grading system. Final grades are assigned from the weighted percentage earned across the four components above, and scores are not rounded up across a boundary.

A+	A	A–	B+	B	B–	C+	C	C–	D+	D	D–	F
97	93	90	87	83	80	77	73	70	67	63	60	<60

Each figure is the minimum percentage for that grade.

All scores post to the ELMS gradebook, normally within one week of the due date. Graded work is returned through ELMS. To review a graded item with the person who scored it, come to office hours within two weeks of the posting. The final exam is retained by the instructor, and you may review it in my office by appointment during the following semester.

In-Class Activities

To reinforce the concepts and encourage you to attempt the reading, activities and exercises will be given during class time throughout the semester. These are open-book individual exercises, completed during the session and submitted before you leave.

The course includes 27 in-class activities. The 5 lowest scores are dropped, so 22 count toward the activity grade. **The drops are automatic and require no notice, documentation, or explanation, and they absorb any cause.** Because the drops already serve as the accommodation, any absence past the fifth reduces the grade regardless of its reason, except as provided below. **A submission recorded while you are not present in class receives a 0 and is treated as an integrity violation.**

A documented extended absence is handled separately: the affected sessions are excluded from the grade and are not absorbed by the drops. Four categories qualify: religious observance, University-sanctioned activities such as athletic or academic travel, military service, and an illness or emergency that prevents attendance for more than one calendar week. A shorter illness is covered by the drops and does not qualify. Notify me through the ELMS Inbox as soon as you are aware of the conflict and no later than 3 days after the first affected session, and provide the documentation the University requires for that category. Approved sessions are removed from the denominator. If more than 8 sessions are excluded, we will discuss an alternative arrangement.

Individual Assignments

There will be five homework assignments to be completed individually. It is not wrong to seek clarifications and minor help with completion of each assignment but there is a very fine line between seeking assistance and cheating. Please stay on the legal side of that line. You are always welcome to the office hours should you have questions.

To prevent plagiarism in homework assignments, the grading of homework may also include an oral part. For each homework assignment, I reserve the right to randomly select a subset of students to come to my office hours and explain the logic of the code that was submitted. If a student cannot clearly explain how own code works, it is up to my discretion to decide whether plagiarism occurs, and corresponding measures will be taken.

Each assignment will require you to spend time on the computer. Please plan on it. The five assignments carry equal weight, and each is worth 4% of the course grade. Assignments post on a Wednesday and are due roughly two weeks later, on the date given in the schedule, by 11:59pm. ELMS permits multiple submissions, but only the latest submission will be graded. Late work is governed by the late assignment policy below.

Course Project

The course project gives you an opportunity to combine the skills you have acquired in the course. You build it alone, and you build it across the whole semester, adding one layer per unit. There is no group project and no team deliverable.

You choose one of two tracks in your project brief. Track A is a repetitive task you personally do, automated. Track B is the Bayline Athletic case, a financial analysis with the data supplied. Both tracks have the same milestones and the same weights, and each track is graded on its own rubric in the project brief. You may switch from Track A to Track B once, up to Monday October 19.

By Unit 4 a Track A program uses a language model as a working component. The bar is one model call, which classifies text, extracts a field, summarizes, or drafts. Tool calling and agent loops raise the grade. They are not required to pass. The Track A final report carries a section stating what the model does in your program and how you checked it, and the checking plan is what carries the weight. Track B does not require a model call.

Deliverable	Due	Weight	Track A · a task you do	Track B · the Bayline case
Project brief	Mon Sep 28	ungraded	The one task you propose, what you plan to do with it, and your own data file attached	The question and what you plan to do to answer it

Deliverable	Due	Weight	Track A · a task you do	Track B · the Bayline case
Midterm build and report	Mon Oct 26	15%	A program that runs on your real data and produces a summary someone could act on	Revenue and contribution margin by month and by channel, tied to the published close-pack totals
Final project, code and report	Mon Dec 7	15%	Code, data, and a report that folds in the AI layer	Code, data, and a report that answers the question and recommends an action
Course project total		30%		

The project brief is a declaration and carries no weight.

I reserve the right to ask any student to walk me through their own submission in office hours, and a student who cannot explain what they submitted receives no credit for it.

The description of the course project will provide you with more detailed information, and it is posted on ELMS in the first three weeks. All project deliverables are due via ELMS by the date and time indicated on the schedule. All projects are due by 11:59pm on the day due. ELMS permits multiple submissions, but only the latest submission will be graded. Late work is governed by the late assignment policy below.

Late Assignment Policy

This policy governs every homework assignment and every project deliverable. The deduction applies to the score the work would otherwise earn.

Submitted	Deduction
Within 1 hour after the deadline	10%
After 1 hour and within 24 hours after the deadline	50%
More than 24 hours after the deadline	100%

An excused absence extends the deadline instead, as stated under Absences and Late Policy.

Final Exam

The final exam is scheduled by the Registrar during the December 14 to 21 examination period. The date is to be determined by the University, and it is announced on ELMS as soon as the Registrar publishes it. Confirm it on Testudo before you book travel.

The exam will include conceptual questions and code examples. It will primarily test whether you understood concepts covered in the lecture and reading, and it weights understanding and correcting code above writing it from scratch. The examination is on paper, open-note, and closed-device.

No extra time will be provided for late arrivals. If your cell phone or mobile device goes off during an exam you will receive a point reduction of 12.5%.

Exam can only be made-up in the event of documented emergencies. Written permission must be obtained 24 hours before the exam if you cannot attend. In any event, make-up exam is only given at instructor discretion.

COURSE-RELATED POLICIES

Students are responsible for knowing their rights and reviewing all course related policies found at UMD's Office of Undergraduate Studies website, <https://www.ugst.umd.edu/courserelatedpolicies.html>. The primary goal of our discussions is to enhance the collective learning of the class. The following guidelines can help you make effective contributions to our class discussions. Please see the following list of tips for effective and meaningful discussion participation.

Examples of Good Discussion Contributions:

- Comments that report and explain content, opinions, or analysis by stating clear assumptions, critically self-reflecting, citing specific sources, and/or conducting explicit, systematic analysis.
- Comments / questions that provide or seek clarification.
- Comments with good timing as well as good substance providing substantiated claims and evidence and sharing sources where possible.
- Comments that make points clearly.
- Comments that move learning forward by building an argument and/or drawing on other comments without repeating.
- Comments that respectfully articulate a point of agreement/disagreement.

Examples of Ineffective Discussion Contributions:

- Destructive attacks.
- Interrupting peers.

- Monopolization of the discussion.
- Disrespectfully articulating a point of agreement/disagreement.
- Out-of-sequence comments / destroying the flow of class discussion.

RESOURCES & ACCOMMODATIONS

Accessibility and Disability Services

The University of Maryland is committed to creating and maintaining a welcoming and inclusive educational, working, and living environment for people of all abilities. The University of Maryland is also committed to the principle that no qualified individual with a disability shall, based on disability, be excluded from participation in or be denied the benefits of the services, programs, or activities of the University, or be subjected to discrimination. The Accessibility & Disability Service (ADS) provides reasonable accommodations to qualified individuals to provide equal access to services, programs and activities. ADS cannot assist retroactively, so it is generally best to request accommodations several weeks before the semester begins or as soon as a disability becomes known. Any student who needs accommodations should contact me as soon as possible so that I have sufficient time to plan. For assistance in obtaining an accommodation, contact Accessibility and Disability Service at 301-314-7682, or email them at adsfrontdesk@umd.edu. More information and services are available from the UMD Counseling Center.

The final exam is handwritten on paper, and alternative formats are available. Raise this with me early if it affects you.

Student Resources and Services

Taking personal responsibility for your own learning means acknowledging when your performance does not match your goals and doing something about it. I hope you will come talk to me so that I can help you find the right approach to success in this course, and I encourage you to visit UMD's Student Academic Success website to learn more about the wide range of campus resources available to you. In particular, everyone can use some help sharpen their communication skills (and improving their grade) by visiting UMD's Writing Center and schedule an appointment with the campus Writing Center. You should also know there are a wide range of resources to support you with whatever you might need (UMD's Student Resources and Services website may help). If you feel it would be helpful to have someone to talk to, visit UMD's Counseling Center.

Basic Needs Security

If you have difficulty affording groceries or accessing sufficient food to eat every day, or lack a safe and stable place to live, please visit UMD's Division of Student

Affairs website for information about resources the campus offers you and let me know if I can help in any way.

Statement on Diversity and Inclusion

As part of the Smith School's commitment to diversity and inclusion, we recognize the importance of a diverse student body as necessary to a THRIVING environment. We are committed to fostering inclusive and equitable classroom environments. The Robert H. Smith School of Business strives to ensure all members of the Smith community feel welcomed, valued, and proud of every aspect of who they are. Through education, knowledge creation, advocacy, programming, and support, Maryland Smith commits to building an inclusive community that fosters a sense of belonging among all stakeholders. Additional UMD-wide resources are available to you in the Office of Diversity and Inclusion.

I invite you, if you wish, to tell us how you want to be referred to both in terms of your name and your pronouns (he/him, she/her, they/them, etc.). The pronouns someone indicates are not necessarily indicative of their gender identity. Additionally, how you identify in terms of your gender, race, class, sexuality, religion, and dis/ability, among all aspects of your identity, is your choice whether to disclose (e.g., should it come up in classroom conversation about our experiences and perspectives) and should be self-identified, not presumed or imposed. I will do my best to address and refer to all students accordingly, and I ask you to do the same for all your fellow Terps. For more resources and information, visit the LGBTQ+ Equity Center.

Technology Policy

Laptops are required and stay open through most of the session. Please refrain from using cellphones and other electronic devices during class sessions unless we have designated such use as part of a class exercise. During the final exam, all devices are closed.

Class sessions are not recorded, and recording a session without my advance permission is prohibited.

Attendance and Participation

- Given the interactive style of this class, attendance will be crucial to note-taking and thus your performance in this class. Attendance is particularly important also because the in-class activity is completed and submitted during the session.
- Each student is expected to make substantive contributions to the learning experience, and attendance is expected for every session.
- Students with a legitimate reason to miss a live session should communicate in advance with the instructor, except in the case of an emergency.

- Students who miss a live session are responsible for learning what they miss from that session.
- Additionally, students must complete all readings and assignments in a timely manner to fully participate in class.

Participation is graded through the activity submitted during each session, which is 20% of the course grade. There is no separate participation score, and speaking in class is not graded.

Absences and Late Policy

Late assignments require prior permission from the instructor and must be accompanied by a legitimate reason for not meeting a target deadline. The complete university policy on absences can be found [here](#). As per university policy, one missed class with an ELMS notification (preferably before) is acceptable. For absences longer than one class session or more than twice in the semester, documentation is required. For Major Scheduled Graded Events (MSGEs), only University approved absences (religious observance, unforeseen life circumstances out of your control (with documentation), or documented illnesses with appropriate documentation will be accepted.

To report an excused absence, send me an ELMS/Canvas Inbox message, in advance where the absence is foreseeable and within 24 hours otherwise. An excused absence extends a homework or project deadline by the length of the absence. In-class activities have no make-up, because the five dropped scores cover ordinary absences.

Score/Grade Appeals

You must make the appeal in writing. However, score changes are at the discretion of the instructor and may be up or down based upon a complete review of the work in question. It is important to recognize that a grade reflects another person's judgment of your work. In this sense, all grading is subjective. Appealing scores is discouraged. Changing a few points on assignments rarely makes a difference in the final grade. Time is much better spent discussing and clarifying the information content presented in the course. In the event of disputes at the end of the semester, you will be required to produce the testing material in question.

In the case of a grading mistake (i.e., grade is posted incorrectly, grader did not give credit for an item that exists) you must first contact the TA who graded the assignment.

Incomplete Grades Policy

A grade of 'Incomplete' may be awarded to students who have a legitimate reason for needing additional time to complete a course. Legitimate reasons include emergencies or extenuating circumstances that prevent a student from completing the course requirements within the normal time frame. Students must initiate the request for an incomplete prior to the end of the semester. In no case will a grade of incomplete be awarded to someone seeking more time to master the course material to improve grade. However, under all circumstances, an 'I' will be awarded only at the discretion of the instructor.

CRITICAL THINKING STANDARDS

Intellectual excellence is one of the key pillars of the undergraduate program at the Robert H. Smith School of Business. At Smith and in your professional life after Smith, you will need to excel at meeting the six critical thinking standards, which are hallmarks of rigorous and evidence-based decision making.

In this course, I will expect you to strive for excellence in meeting these six standards in your contributions to class discussions and your written work. You should expect me and your classmates to push you to meet these standards. For example, in a class discussion, please fully explain your point and how it relates to the question that was posed to the class. Describe the evidence on which you base your claims. Be open to feedback about where you can do more to meet these standards.

ACADEMIC INTEGRITY AND MISCONDUCT

The University's Code of Academic Integrity is designed to ensure that the principles of academic honesty and integrity are upheld. In accordance with this code, the Smith School does not tolerate academic dishonesty. Please ensure that you fully understand this code and its implications because all acts of academic dishonesty will be dealt with in accordance with the provisions of this code. All students are expected to adhere to this Code. It is your responsibility to read it and know what it says, so you can start your professional life on the right path. As future professionals, your commitment to high ethical standards and honesty begins with your time at the Smith School.

It is important to note that course assistance websites, such as CourseHero, Chegg, etc., are not permitted sources for Smith School courses. Material taken or copied from these sites can be deemed unauthorized material and a violation of academic integrity. These sites offer information that might not be accurate and that shortcut the learning process, particularly the critical thinking steps necessary for college-level assignments.

Additionally, it is understandable that students may use a variety of online or virtual forums for course-wide discussion (e.g., GroupMe or WeChat).

Collaboration in this way regarding concepts discussed in this course is permissible. However, collaboration on graded activities and assignments is strictly prohibited. Examples of prohibited collaboration include asking for answers on grading components, asking for access codes to grading components, etc.

On each exam and assignment you write out and sign the following pledge: "I pledge on my honor that I have not given or received any unauthorized assistance on this exam/assignment."

Artificial Intelligence Policy

The permission widens as the semester goes on. Every graded item and every in-class activity carries one of three labels, printed on the assignment or on the session's slide that morning.

The No AI label covers the final exam and the in-class activities in Sessions 1, 5, and 27. You may use your notes and course materials. You may not use any AI assistant, including Copilot, Gemini, ChatGPT, Claude, or the AI features built into Colab and Visual Studio Code.

The AI as tutor label covers all five homework assignments, the course project through Session 21, and the in-class activity in most sessions. You may ask an assistant to explain a concept, describe an error message, review code you already wrote, or generate practice problems. You may not paste an assignment prompt into an assistant and submit what comes back, and you may not submit code you cannot explain line by line.

The AI as collaborator label covers class work and project work from Session 22 onward. Use AI assistants without restriction to write, debug, and refactor code. You remain the owner of every submission.

Every homework and project submission ends with a short disclosure disclosing which tool you used, what you asked it, and what you did with the answer. Using an assistant heavily does not lower the grade, and a missing disclosure is a violation of the Code of Academic Integrity.

To help you avoid unintentional violations, the following table lists levels of collaboration that are acceptable for each type of assignment. If you ever feel pressured to comply with someone else's academic integrity violation, please reach out to me straight away. Also, if you are ever unclear about acceptable levels of collaboration, please ask!

	Open Notes	Use Book	Ask Friends	Search Online	Work in Groups	AI Assistant
In-Class Activity	yes	yes	no	no	no	as the session's slide states
Individual Assignment	yes	yes	cited	cited	no	tutor, disclosed
Course Project, through Session 21	yes	yes	cited	cited	no	tutor, disclosed
Course Project, from Session 22	yes	yes	cited	cited	no	unrestricted, cited
Final Exam	yes	yes	no	no	no	no

Tips for Success

- **Participate:** Discussions are a critical part of the course. You can learn as much, if not more, from the ideas and perspectives of your peers and instructor. Participation can also help you articulate your ideas and develop critical thinking skills.
- **Manage your time:** Make time for your learning and participation each week. Give yourself plenty of time to complete assignments (plus extra time to handle any technology problems). A 3-credit course assumes roughly six hours of work per week beyond the meeting time, and short daily practice works better than one long session.
- **Login regularly:** Log in to ELMS several times a week to view announcements. In-class materials and any change of plan appear there first.
- **Try not to fall behind:** This class moves at a quick pace and each week builds on the previous. It is very difficult to catch up if you wait until the last minute to complete your assignments or skip a learning unit. Please reach out to me early if you feel you're slipping behind on the content and I will try to help.
- **Ask for help if needed:** If you need help with ELMS or other technology, contact Smith IT or email OTL-EdTech@umd.edu. If you are struggling with a course concept, reach out to your peers, TAs, and me for support. If you have been stuck for more than an hour, ask.

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Course materials are copyrighted and may not be reproduced for anything other than personal use without written permission. Although every effort has been made to be complete and accurate, unforeseen circumstances arising during the semester could require the adjustment of material given here. Consequently, given notice to students, the instructor reserves the right to change any information on this syllabus or in other course materials.

(end of syllabus)

BMGT 302 Essential Programming and AI Skills for Business Analytics

Fall 2026 Section 0401 – Tentative Course Schedule

Week	Date	Topic	Reading	Due
1	Mon 8/31	Syllabus, and what programming is for in the era of generative AI. Install Python and VS Code	Syllabus	
	Wed 9/2	Using a language model to learn programming	Syllabus, AI policy	
2	Mon 9/7	<i>Labor Day, no class</i>		
	Wed 9/9	Decomposition and algorithms	HW1 brief	HW1 Post: Wed. Sep. 9
3	Mon 9/14	Variables, types, expressions	ELMS	
	Wed 9/16	Conditionals and business rules	ELMS	
4	Mon 9/21	Loops and repetition	ELMS	HW1 Due: Mon. Sep. 21 @11:59pm
	Wed 9/23	Lists, strings, and debugging	Python tutorial: errors	HW2 Post: Wed. Sep. 23
5	Mon 9/28	Dictionaries, and choosing a data structure	ELMS	Project Brief Due: Mon. Sep. 28 @11:59pm
	Wed 9/30	Functions and decomposition	ELMS	
6	Mon 10/5	Files, CSV, and error handling	Python tutorial: exceptions	HW2 Due: Mon. Oct. 5 @11:59pm

Week	Date	Topic	Reading	Due
	Wed 10/7	Objects	Python docs: text sequences	HW3 Post: Wed. Oct. 7
7	Mon 10/12	<i>Fall Break, no class</i>		
	Wed 10/14	Classes	ELMS	
8	Mon 10/19	pandas package	10 Minutes to pandas	Last day to switch Track A to Track B
	Wed 10/21	Selecting, filtering, and cleaning	ELMS	HW3 Due: Wed. Oct. 21 @11:59pm
9	Mon 10/26	Grouping and joining	pandas merge	Midterm Build and Report Due: Mon. Oct. 26 @11:59pm
	Wed 10/28	Visualization for business audiences	pandas DataFrame.plot	HW4 Post: Wed. Oct. 28
10	Mon 11/2	Reading and writing Excel workbooks from Python	pandas read_excel	
	Wed 11/4	VBA: machine- generated code, and choosing the tool	VBA documentation	
11	Mon 11/9	How language models work	ELMS	HW4 Due: Mon. Nov. 9 @11:59pm
	Wed 11/11	Prompt engineering for code	ELMS	HW5 Post: Wed. Nov. 11
12	Mon 11/16	AI coding assistants in the loop, and verification discipline	ELMS	
	Wed 11/18	AI-augmented programming: code quality, refactoring, version control	ELMS	<i>Collaborator tier opens</i>

Week	Date	Topic	Reading	Due
13	Mon 11/23	Calling a language model from Python: classify, extract, summarize	ELMS	
	Wed 11/25	<i>Thanksgiving recess, no class</i>		
14	Mon 11/30	From analysis to a deck: generating slides and documents with AI	ELMS	HW5 Due: Mon. Nov. 30 @11:59pm
	Wed 12/2	Agentic AI: Skills	ELMS	
15	Mon 12/7	Agentic AI: file access, control, edit	ELMS	Final Project Due: Mon. Dec. 7 @11:59pm
	Wed 12/9	Review for the final exam	ELMS	
Final	Dec 14-21	Final Exam Section 0401		Date and room set by the Registrar, TBD

Every session carries an in-class activity, scored 0 or 1 for attendance, with the 5 lowest dropped. Readings for each session are posted on ELMS. This schedule is tentative and is subject to change as necessary; monitor the course ELMS page for current deadlines. In the unlikely event of a prolonged university closing, or an extended absence from the university, adjustments to the schedule, the deadlines, and the assignments will be made based on the duration of the closing and the specific dates missed.